CS351 - IT Workshop II (Cloud Computing)

Assignment 5

Total Marks: 40

Please follow the instructions below. Marks will be deducted if you do not follow these instructions.

(i) Screenshots: Take clear screenshots, resize the images to reduce their file size, and upload them to Google Classroom. Please do not upload a .zip file in Google Classroom.

(ii) Source Code File Naming: Write different Python files for different questions. Name each Python source file using the following format: <assignment_no>_<question_no>.py

For Example: a1_q1.py

(iii) Source Code Header: Include your name and roll number at the top of each source file using the following format.

Name: <Your Name>

Roll No: <Your Roll No>

(iv) Proper Comments: You need to write comments for each block of code, function inputs and outputs, and the working logic of the functions.

(v) Proper Variable and Function Names: Please use meaningful variable and function names.

Part 1: Your Website and Startup Script [10 Marks]

You are required to ***set up and host a new website*** on an AWS EC2 instance.

- Create the website with at least two HTML pages and CSS/images.
- Write a startup script that:
    - Installs and starts the Apache HTTP server.
    - Copies the website files to the Apache document root (/var/www/html).

    - Displays the EC2 instance ID on the homepage (using instance metadata).

Part 2: Automation using boto3 **[10 Marks]**

Automate the ***Website (Part 1)*** Deployment using Python and boto3.

- Write a Python script that:
    - Creates a Security Group that allows inbound HTTP traffic on port 80.
    - Launches an EC2 instance using:
        - The startup script from Part 1.
        - A specified AMI (Ubuntu LTS).
        - An t3.micro instance type.
    - Attaches the Security Group to the instance.
- After launching, retrieve and print:
    - The instance ID.
    - The instance Public DNS name.
- Test your script by opening the public DNS in a browser and confirming the website works.

Part 3: Auto Scaling with CloudWatch **[20 Marks]**

Extend your automation to create an ***Auto Scaling Group (ASG)*** with ***CloudWatch alarms*** and ***scaling policies***.

- **Modify your Python script** to:
    - Create a ***Launch Template*** or ***Launch Configuration*** that uses:
        - The same AMI, instance type, Security Group, and startup script *from previous parts*.
    - Create an ***Auto Scaling Group*** in an Availability Zone (i.e. Mumbai (ap-south-1)) with:
        - Min size: 1, Max size: 3, Desired capacity: 1.
    - Configure ***scale-out*** and ***scale-in*** policies:

        - **Scale Up:** If ***average CPU ≥ 20% for 2 minutes*** → add 1 instance.
        - **Scale Down:** If ***average CPU ≤ 30% for 2 minutes*** → remove 1 instance.
    - Create ***CloudWatch alarms*** to call these policies.
- ***Test*** auto-scaling by:
    - ***Applying load*** (e.g., ***Apache Benchmark*** or ***stress***) to trigger *scale-up*.
    - ***Removing load*** to trigger *scale-down*.
- ***Verify*** by checking:
    - ***Multiple EC2 instances*** serving the site (confirm by refreshing and checking the ***instance ID*** on the page).
    - ***CloudWatch alarms*** transitioning between states.